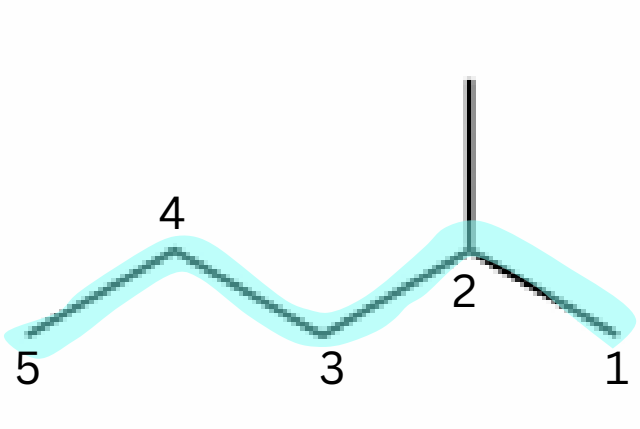
AO1 FOUNDATION - C_6H_{14}

PART A - STRUCTURE → NAME

1. NUMBERED PARENT



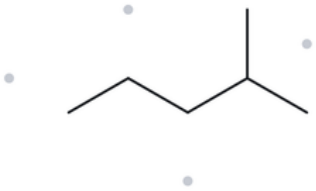
2. NAMING TRAIL

Parent: longest chain = 5 C → pentane. Several 5-C paths exist, but each carries one methyl group, so they just lead to the same name.
Numbering: left → right gives 4; right → left gives 2. First point of difference: 2 < 4, so number from the right.
Substituents: methyl on C2 → 2-methyl.

3. MY NAME

2-methylpentane

4. THE ALKANE CHECK



2-methylpentane

MATCH

VERIFICATION STRIP

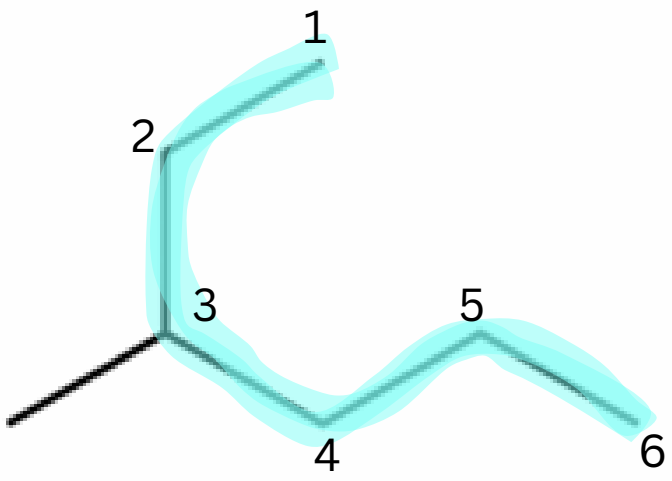
☑ LONGEST CHAIN CONFIRMED ☑ LOWEST LOCANTS

☑ ALPHABETICAL ORDER ☑ PUNCTUATIONS ☑ FORMULA COUNTED

AO2 FOUNDATION - C_7H_{16}

PART A - STRUCTURE → NAME

1. NUMBERED PARENT



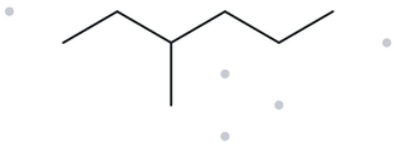
2. NAMING TRAIL

Parent: longest chain = 6 C → hexane. Only one 6-C path exists.
Numbering: left → right gives 3; right → left gives 4. First point of difference: 3 < 4, so number from the left.
Substituents: methyl on C3 → 3-methyl.

3. MY NAME

3-methylhexane

4. THE ALKANE CHECK



MATCH

3-methylhexane

VERIFICATION STRIP

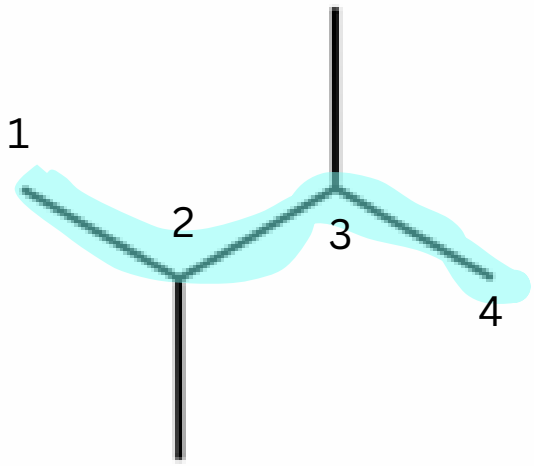
☐ LONGEST CHAIN CONFIRMED ☐ LOWEST LOCANTS

☐ ALPHABETICAL ORDER ☐ PUNCTUATIONS ☐ FORMULA COUNTED

AO3 FOUNDATION - C_6H_{14}

PART A - STRUCTURE → NAME

1. NUMBERED PARENT



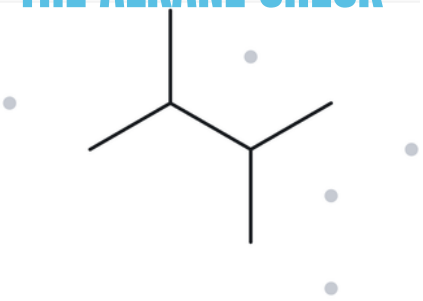
2. NAMING TRAIL

Parent: longest chain = 4 C → butane. Several 4-C paths exist, but each carries two methyl groups, so they lead to the same name.
Numbering: left → right gives 2,3; right → left gives 2,3. The locant sets are identical, so either direction works.
Substituents: methyl on C2 and C3 → 2,3-dimethyl.

3. MY NAME

2,3-dimethylbutane

4. THE ALKANE CHECK



MATCH

2,3-dimethylbutane

VERIFICATION STRIP

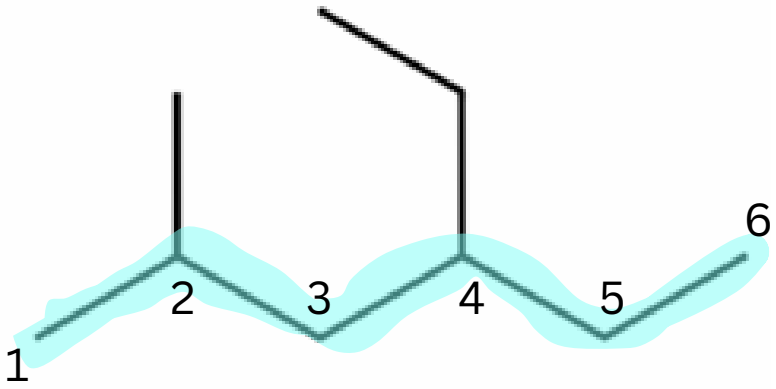
☐ LONGEST CHAIN CONFIRMED ☐ LOWEST LOCANTS

☐ ALPHABETICAL ORDER ☐ PUNCTUATIONS ☐ FORMULA COUNTED

AO4 DEVELOPING - C_9H_{20}

PART A - STRUCTURE → NAME

1. NUMBERED PARENT



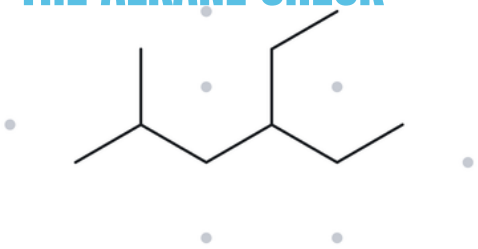
2. NAMING TRAIL

Parent: longest chain = 6 C → hexane. Several 6-C paths exist, but each carries one methyl and one ethyl, so they lead to the same name. **Numbering:** left → right gives 2,4; right → left gives 3,5. First point of difference: 2 < 3, so number from the left.
Substituents: methyl on C2, ethyl on C4 (listed alphabetically) → 4-ethyl-2-methyl.

3. MY NAME

4-ethyl-2-methylhexane

4. THE ALKANE CHECK



MATCH

4-ethyl-2-methylhexane

VERIFICATION STRIP

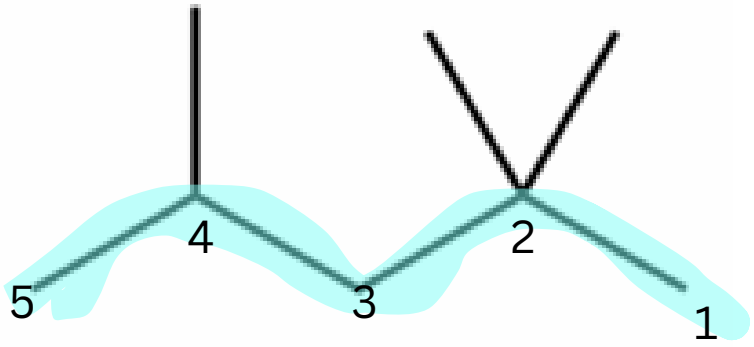
☑ LONGEST CHAIN CONFIRMED ☑ LOWEST LOCANTS

☑ ALPHABETICAL ORDER ☑ PUNCTUATIONS ☑ FORMULA COUNTED

AO5 DEVELOPING - C_8H_{18}

PART A - STRUCTURE → NAME

1. NUMBERED PARENT



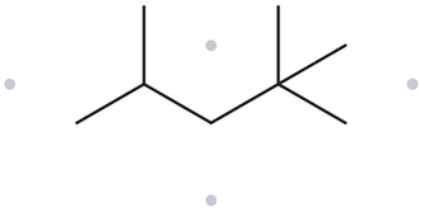
2. NAMING TRAIL

Parent: longest chain = 5 C → pentane. Several 5-C paths exist, but each carries three methyl groups, so they lead to the same name.
Numbering: left → right gives 2,4,4; right → left gives 2,2,4. First point of difference: at the second locant, 2 < 4, so number from the right (the end with the two methyls).
Substituents: methyl on C2 (2) and C4 → 2,2,4-trimethyl.

3. MY NAME

2,2,4-trimethylpentane

4. THE ALKANE CHECK



MATCH

2,2,4-trimethylpentane

VERIFICATION STRIP

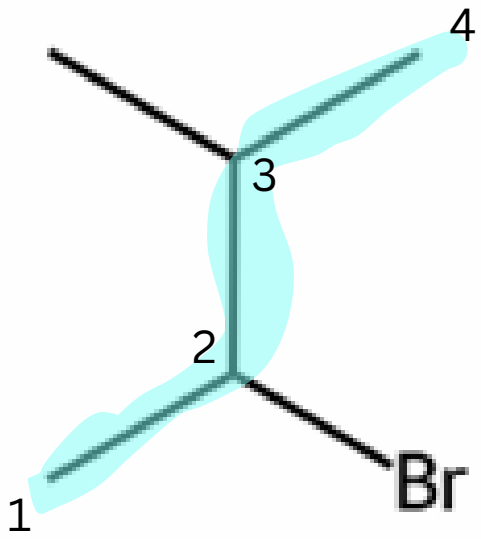
☐ LONGEST CHAIN CONFIRMED ☐ LOWEST LOCANTS

☐ ALPHABETICAL ORDER ☐ PUNCTUATIONS ☐ FORMULA COUNTED

AO6 DEVELOPING - $C_5H_{11}Br$

PART A - STRUCTURE → NAME

1. NUMBERED PARENT



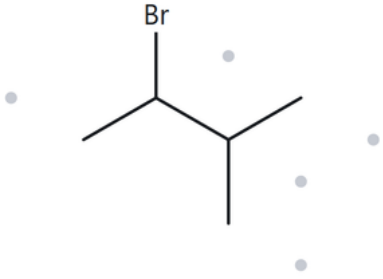
2. NAMING TRAIL

Parent: longest chain = 4 C → butane. Several 4-C paths exist, but each carries one bromo and one methyl, so they lead to the same name.
Numbering: both directions give 2,3. The locant sets are identical, so the lower number goes to the substituent that comes first alphabetically (bromo before methyl). Number from the bromo end.
Substituents: bromo on C2, methyl on C3 (listed alphabetically) → 2-bromo-3-methyl.

3. MY NAME

2-bromo-3-methylbutane

4. THE ALKANE CHECK



MATCH

2-bromo-3-methylbutane

VERIFICATION STRIP

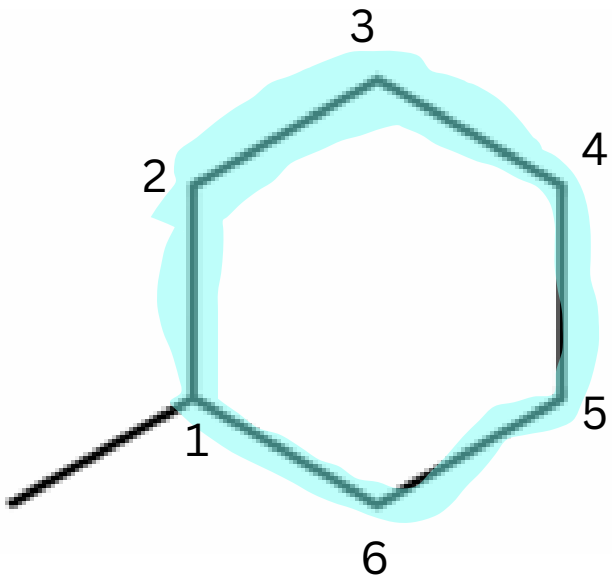
☐ LONGEST CHAIN CONFIRMED ☐ LOWEST LOCANTS

☐ ALPHABETICAL ORDER ☐ PUNCTUATIONS ☐ FORMULA COUNTED

AO7 DEVELOPING - C_7H_{14}

PART A - STRUCTURE → NAME

1. NUMBERED PARENT



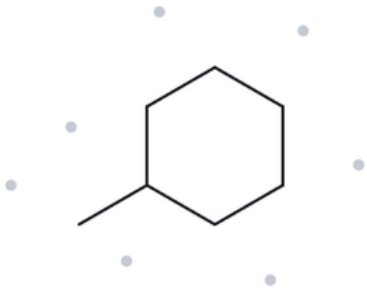
2. NAMING TRAIL

Parent: ring size = 6 C vs substituent = 1 C. The ring is larger → cyclohexane.
Numbering: the single substituent takes position 1, and the direction doesn't matter because there is nothing else to compare.
Substituents: methyl on C1 (locant 1 is omitted for a single ring substituent) → methyl.

3. MY NAME

methylcyclohexane

4. THE ALKANE CHECK



MATCH

methylcyclohexane

VERIFICATION STRIP

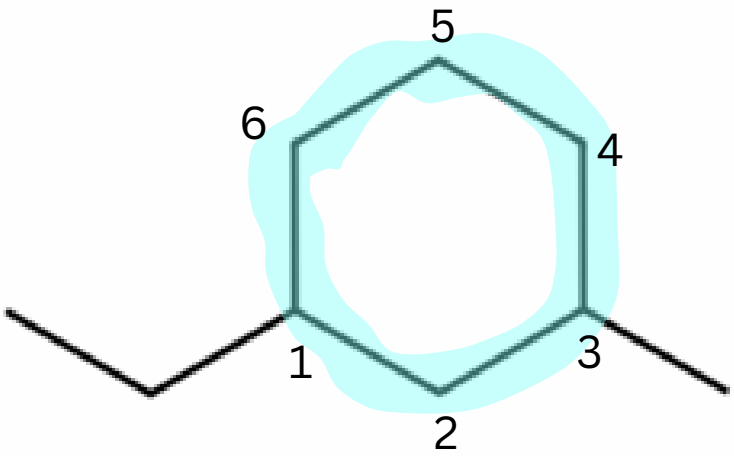
☐ LONGEST CHAIN CONFIRMED ☐ LOWEST LOCANTS

☐ ALPHABETICAL ORDER ☐ PUNCTUATIONS ☐ FORMULA COUNTED

AO8 CHALLENGE - C_9H_{18}

PART A - STRUCTURE → NAME

1. NUMBERED PARENT



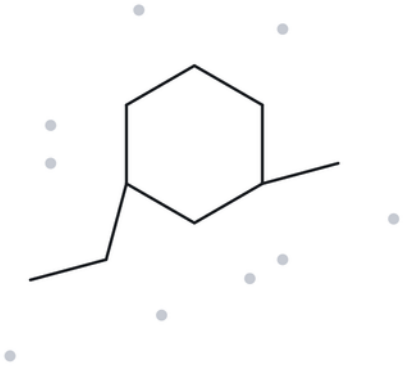
2. NAMING TRAIL

Parent: ring size = 6 C vs largest substituent = 2 C. The ring is larger → cyclohexane.
Numbering: going the short way around between the two substituents gives 1,3; the long way gives 1,5. Lowest locant set: 1,3. The locant sets tie from either substituent, so C1 goes to the first one alphabetically (ethyl before methyl).
Substituents: ethyl on C1, methyl on C3 (listed alphabetically) → 1-ethyl-3-methyl.

3. MY NAME

1-ethyl-3-methylcyclohexane

4. THE ALKANE CHECK



MATCH

1-ethyl-3-methylcyclohexane

VERIFICATION STRIP

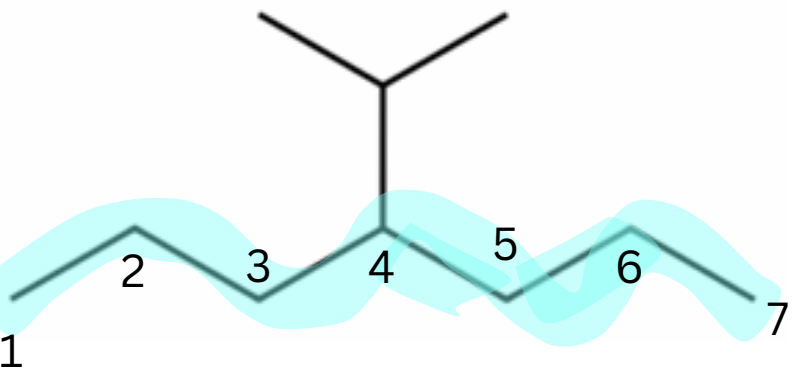
☑ LONGEST CHAIN CONFIRMED ☑ LOWEST LOCANTS

☑ ALPHABETICAL ORDER ☑ PUNCTUATIONS ☑ FORMULA COUNTED

AO9 CHALLENGE - $C_{10}H_{22}$

PART A - STRUCTURE → NAME

1. NUMBERED PARENT



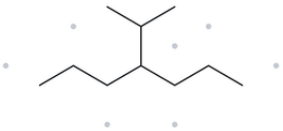
2. NAMING TRAIL

Parent: longest chain = 7 C → heptane. The chain runs through both propyl arms. The path through the isopropyl group is only 6 C.
Numbering: left → right gives 4; right → left gives 4. The chain is symmetrical about C4, so either direction works.
Substituents: isopropyl on C4 → 4-isopropyl

3. MY NAME

4-isopropylheptane

4. THE ALKANE CHECK



2-methyl-3-propylhexane

MISMATCH

The name must be based on the longest continuous carbon chain, which is 7 for hepta. This makes the site's six-carbon path incorrect.

VERIFICATION STRIP

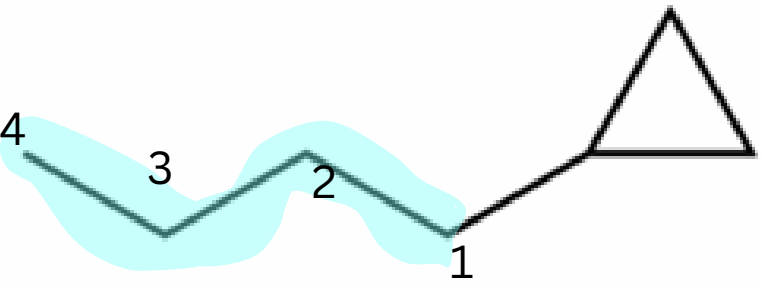
☒ LONGEST CHAIN CONFIRMED ☒ LOWEST LOCANTS

☒ ALPHABETICAL ORDER ☒ PUNCTUATIONS ☒ FORMULA COUNTED

A10 CHALLENGE - C_7H_{14}

PART A - STRUCTURE → NAME

1. NUMBERED PARENT



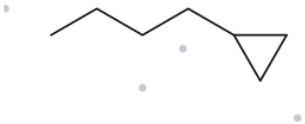
2. NAMING TRAIL

Parent: open chain vs ring = 4 C vs 3 C. Since 4 > 3, the chain is the parent → butane.
Numbering: right → left gives 1; left → right gives 4. First point of difference: 1 < 4, so number from the end attached to the ring.
Substituents: cyclopropyl on C1 → 1-cyclopropyl.

3. MY NAME

1-cyclopropylbutane

4. THE ALKANE CHECK



butylcyclopropane

✗ MISMATCH

The four-carbon chain will take priority as the parent rather than the enclosed ring that only comprises of 3 carbons.

VERIFICATION STRIP

☒ LONGEST CHAIN CONFIRMED ☒ LOWEST LOCANTS

☒ ALPHABETICAL ORDER ☒ PUNCTUATIONS ☒ FORMULA COUNTED

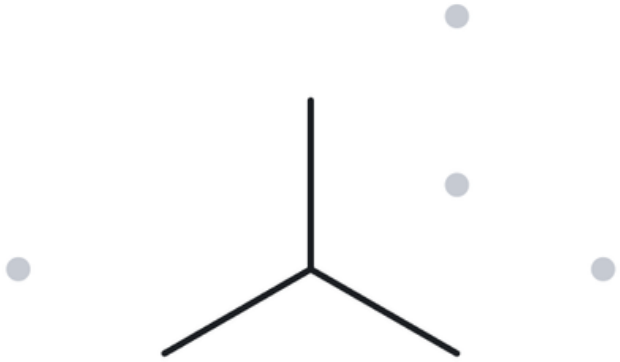
BO1 FOUNDATION - 2-methylpropane

PART B - NAME → STRUCTURE

1. NAME DECODED

Parent: propane → a 3-carbon chain.
Substituents: methyl on C2.
Expected Formula: 3 + 1 = 4 C; open chain → C_4H_{10}

2. BOND-LINE STRUCTURE (THE ALKANE)



3. FORMULA CHECK

Counted from the drawing: C_4H_{10}.
Open chain: C_nH_{2n+2} = C_4H_{10} ✓

4. BACK-CHECK

Returned by The Alkane: 2-methylpropane.

Same as the given name

VERIFICATION STRIP

- ☐ PARENT LENGTH MATCHES
- ☐ SUBSTITUENT POSITIONS MATCH
- ☐ FORMULA AGREES
- ☐ BACK-CHECK RECORDED

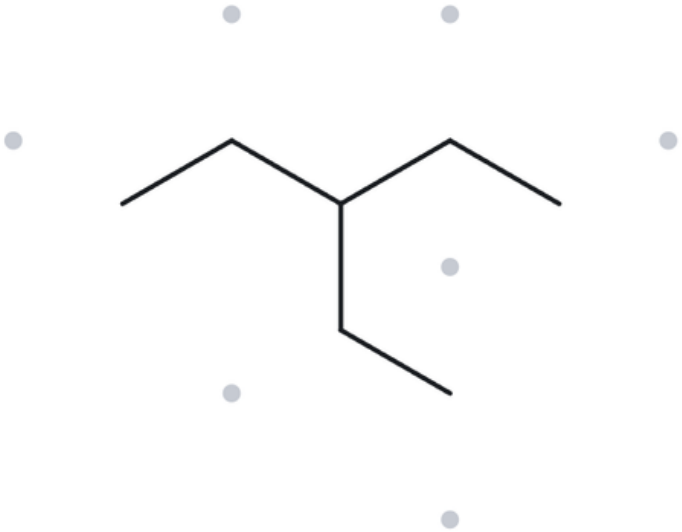
BO2 FOUNDATION - 3-ethylpentane

PART B - NAME → STRUCTURE

1. NAME DECODED

Parent: pentane → a 5-carbon chain.
Substituents: ethyl on C3.
Expected formula: 5 + 2 = 7 C; open chain → C_7H_{16}

2. BOND-LINE STRUCTURE (THE ALKANE)



3. FORMULA CHECK

Counted from the drawing: C_7H_{16}.
Open chain: C_nH_{2n+2} = C_7H_{16} ✓

4. BACK-CHECK

Returned by The Alkane: 3-ethylpentane.

Same as the given name

VERIFICATION STRIP

- ☐ PARENT LENGTH MATCHES
- ☐ SUBSTITUENT POSITIONS MATCH
- ☐ FORMULA AGREES
- ☐ BACK-CHECK RECORDED

BO3 FOUNDATION - 2,2-dimethylbutane

PART B - NAME → STRUCTURE

1. NAME DECODED

Parent: butane → a 4-carbon chain.
Substituents: methyl on C2 (×2).
Expected formula: 4 + 2 = 6 C; open chain → C_6H_{14}

2. BOND-LINE STRUCTURE (THE ALKANE)



3. FORMULA CHECK

Counted from the drawing: C_6H_{14}.
Open chain: C_nH_{2n+2} = C_6H_{14} ✓

4. BACK-CHECK

Returned by The Alkane: 2,2-dimethylbutane.

Same as the given name

VERIFICATION STRIP

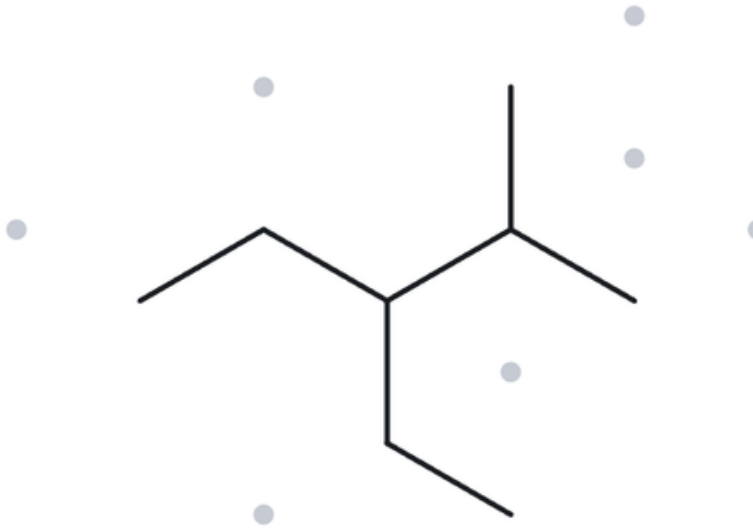
- ☑ PARENT LENGTH MATCHES
- ☑ SUBSTITUENT POSITIONS MATCH
- ☑ FORMULA AGREES
- ☑ BACK-CHECK RECORDED

BO4 DEVELOPING -
3-ethyl-2 methylpentane
PART B - NAME → STRUCTURE

1. NAME DECODED

Parent: pentane → a 5-carbon chain.
Substituents: ethyl on C3; methyl on C2.
Expected formula: 5 + 2 + 1 = 8 C; open chain → C_8H_{18}

2. BOND-LINE STRUCTURE (THE ALKANE)



3. FORMULA CHECK

Counted from the drawing: C_8H_{18}.
Open chain: C_nH_{2n+2} = C_8H_{18} ✓

4. BACK-CHECK

Returned by The Alkane: 3-ethyl-2-methylpentane.

Same as the given name

VERIFICATION STRIP

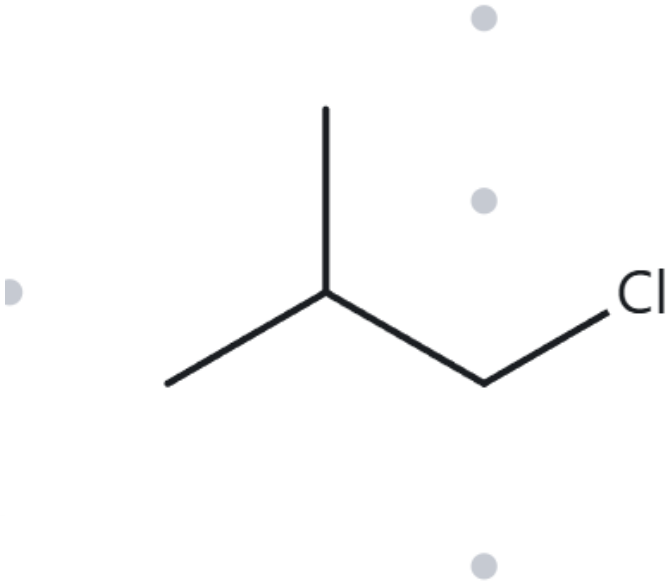
- ☐ PARENT LENGTH MATCHES
- ☐ SUBSTITUENT POSITIONS MATCH
- ☐ FORMULA AGREES
- ☐ BACK-CHECK RECORDED

BO5 DEVELOPING -
1-chloro-2 methylpropane
PART B - NAME → STRUCTURE

1. NAME DECODED

Parent: propane → a 3-carbon chain.
Substituents: chloro on C1; methyl on C2.
Expected formula: 3 + 1 = 4 C; open chain → C_4H_{10}; Cl replaces one H → C_4H_9Cl

2. BOND-LINE STRUCTURE (THE ALKANE)



3. FORMULA CHECK

Counted from the drawing: C_4H_9Cl.
Open chain: C_nH_{2n+2} = C_4H_{10}; minus one H for Cl = C_4H_9Cl ✓

4. BACK-CHECK

Returned by The Alkane: 1-chloro-2-methylpropane.

Same as the given name

VERIFICATION STRIP

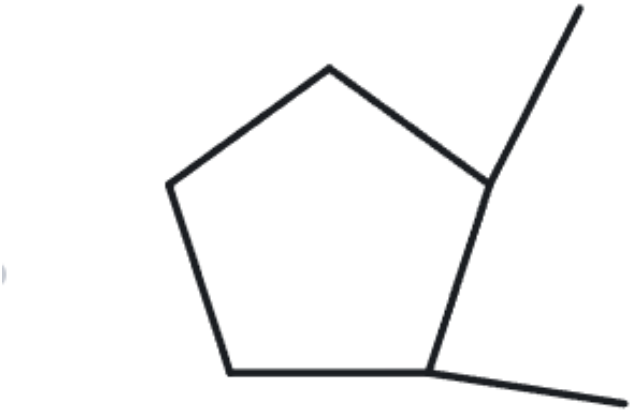
- ☐ PARENT LENGTH MATCHES
- ☐ SUBSTITUENT POSITIONS MATCH
- ☐ FORMULA AGREES
- ☐ BACK-CHECK RECORDED

BO6 DEVELOPING -
1,2 dimethylcyclopentane
PART B - NAME → STRUCTURE

1. NAME DECODED

Parent: cyclopentane → a 5-carbon ring. **Substituents:** methyl on C1; methyl on C2.
Expected formula: 5 + 1 + 1 = 7 C; one ring → C_7H_{14}

2. BOND-LINE STRUCTURE (THE ALKANE)



3. FORMULA CHECK

Counted from the drawing: C_7H_{14}.
One ring: C_nH_{2n} = C_7H_{14} ✓

4. BACK-CHECK

Returned by The Alkane: 1,2-dimethylcyclopentane.

Same as the given name

VERIFICATION STRIP

- ☐ PARENT LENGTH MATCHES
- ☐ SUBSTITUENT POSITIONS MATCH
- ☐ FORMULA AGREES
- ☐ BACK-CHECK RECORDED

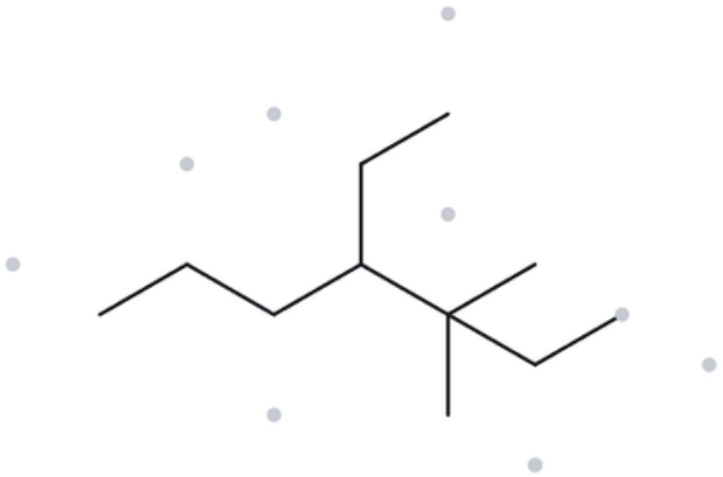
BO7 DEVELOPING - 4-ethyl-3,3 dimethylheptane

PART B - NAME → STRUCTURE

1. NAME DECODED

Parent: heptane → a 7-carbon chain.
Substituents: ethyl on C4; methyl on C3 (×2).
Expected formula: 7 + 2 + 1 + 1 = 11 C; open chain → $C_{11}H_{24}$

2. BOND-LINE STRUCTURE (THE ALKANE)



3. FORMULA CHECK

Counted from the drawing: $C_{11}H_{24}$.
Open chain: C_nH_{2n+2} = $C_{11}H_{24}$ ✓

4. BACK-CHECK

Returned by The Alkane: 4-ethyl-3,3-dimethylheptane.

Same as the given name

VERIFICATION STRIP

- [x] PARENT LENGTH MATCHES
- [x] SUBSTITUENT POSITIONS MATCH
- [x] FORMULA AGREES
- [x] BACK-CHECK RECORDED

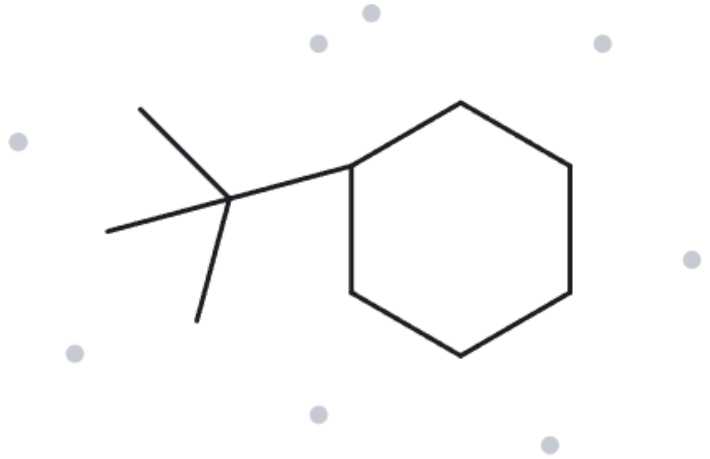
BO8 CHALLENGE- tert-butylcyclohexane

PART B - NAME → STRUCTURE

1. NAME DECODED

Parent: cyclohexane → a 6-carbon ring.
Substituents: tert-butyl on C1.
Expected formula: 6 + 4 = 10 C; one ring → $C_{10}H_{20}$

2. BOND-LINE STRUCTURE (THE ALKANE)



3. FORMULA CHECK

Counted from the drawing: $C_{10}H_{20}$.
One ring: C_nH_{2n} = $C_{10}H_{20}$ ✓

4. BACK-CHECK

Returned by The Alkane: (1,1-dimethylethyl)cyclohexane.

⊗ NOT THE SAME AS THE GIVEN NAME

According to the IUPAC naming reference (Table B.3 on appendix B), naming must prioritize systematic over the common. In this case, (1,1-dimethylethyl)cyclohexane should be the name rather than the given tert-butylcyclohexane.

VERIFICATION STRIP

☒ PARENT LENGTH MATCHES ☒ SUBSTITUENT POSITIONS MATCH

☒ FORMULA AGREES ☒ BACK-CHECK RECORDED

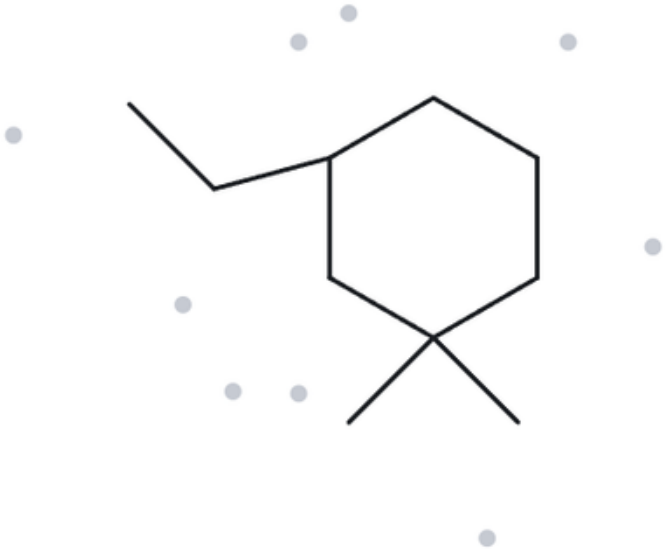
BO9 CHALLENGE - 3-ethyl-1,1-dimethylcyclohexane

PART B - NAME → STRUCTURE

1. NAME DECODED

Parent: cyclohexane → a 6-carbon ring.
Substituents: ethyl on C3; methyl on C1 (2).
Expected formula: 6 + 2 + 1 + 1 = 10 C; one ring → $C_{10}H_{20}$

2. BOND-LINE STRUCTURE (THE ALKANE)



3. FORMULA CHECK

Counted from the drawing: $C_{10}H_{20}$.
One ring: C_nH_{2n} = $C_{10}H_{20}$ ✓

4. BACK-CHECK

Returned by The Alkane: 3-ethyl-1,1-dimethylcyclohexane.

Same as the given name

VERIFICATION STRIP

- ☒ PARENT LENGTH MATCHES
- ☒ SUBSTITUENT POSITIONS MATCH
- ☒ FORMULA AGREES
- ☒ BACK-CHECK RECORDED

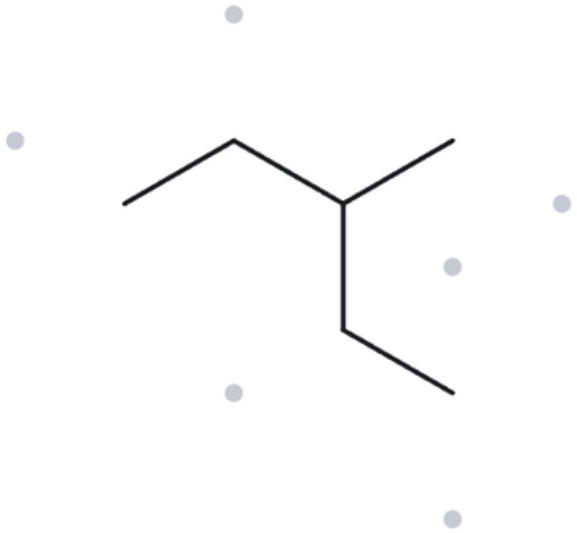
B10 CHALLENGE - 2-ethylbutane

PART B - NAME → STRUCTURE

1. NAME DECODED

Parent: butane → a 4-carbon chain.
Substituents: ethyl on C2.
Expected formula: 4 + 2 = 6 C; open chain → C_6H_{14}

2. BOND-LINE STRUCTURE (THE ALKANE)



3. FORMULA CHECK

Counted from the drawing: C_6H_{14}.
Open chain: C_nH_{2n+2} = C_6H_{14} ✓

4. BACK-CHECK

Returned by The Alkane: 3-methylpentane. .

NOT THE SAME AS THE GIVEN NAME

The chain through the ethyl group gives 5 carbons (pentane) with a methyl on C3, so the longest-chain rule makes 3-methylpentane the correct name.

VERIFICATION STRIP

- ☒ PARENT LENGTH MATCHES
- ☒ SUBSTITUENT POSITIONS MATCH
- ☒ FORMULA AGREES
- ☒ BACK-CHECK RECORDED